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SECONDARY BATTERY AND BATTERY PACK

Abstract

A secondary battery is provided and includes an electrode wound body includes a first end face and a second end face. The first end face faces a positive electrode current collector plate in a height direction. The second end face faces a negative electrode current collector plate in the height direction. A positive electrode includes a positive electrode current collector covered region and a positive electrode current collector exposed region. The positive electrode current collector covered region is a region in which a positive electrode current collector is covered with a positive electrode active material layer. The positive electrode current collector exposed region is a region in which the positive electrode current collector is exposed without being covered with the positive electrode active material layer. All or a part of the positive electrode current collector exposed region forms the first end face and is coupled to the positive electrode current collector plate. A negative electrode includes a negative electrode current collector covered region and a negative electrode current collector exposed region. The negative electrode current collector covered region is a region in which a negative electrode current collector is covered with a negative electrode active material layer. The negative electrode current collector exposed region is a region in which the negative electrode current collector is exposed without being covered with the negative electrode active material layer. All or a part of the negative electrode current collector exposed region forms the second end face and is coupled to the negative electrode current collector plate. The positive electrode includes an insulating layer that covers an opposing part, of the positive electrode current collector exposed region, that is opposed to the negative electrode active material layer with a separator interposed therebetween. The insulating layer has an elongation of 180 percent or higher.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] The present application is a continuation of International Application No. PCT/JP2024/009749, filed on Mar. 13, 2024, which claims priority to Japanese Patent Application No. 2023-056584, filed on Mar. 30, 2023, the entire contents of which are incorporated herein by reference.

BACKGROUND
[0002] The present disclosure relates to a secondary battery, and to a battery pack including the secondary battery.
[0003] Various kinds of electronic equipment, including mobile phones, have been widely used. Such widespread use has promoted development of a secondary battery as a power source that is smaller in size and lighter in weight and allows for a higher energy density. The secondary battery includes a battery device contained inside an outer package member. A configuration of the secondary battery has been considered in various ways.
[0004] A secondary battery is proposed in which what is called a tabless structure is employed to thereby reduce an internal resistance and to allow for charging and discharging with a relatively large current.
SUMMARY
[0005] The present disclosure relates to a secondary battery, and to a battery pack including the secondary battery.
[0006] Consideration has been given in various ways to improve performance of a secondary battery. However, there is still room for improvement in safety of the secondary battery.
[0007] It is therefore desirable to provide a secondary battery that is superior in safety.
[0008] A secondary battery according to an embodiment of the present disclosure includes a positive electrode current collector plate, a negative electrode current collector plate, and an electrode wound body. The electrode wound body is disposed between the positive electrode current collector plate and the negative electrode current collector plate. The electrode wound body includes a stacked body and has a through hole. The stacked body includes a positive electrode, a negative electrode, and a separator and is wound. The through hole is provided through the electrode wound body in a height direction. The electrode wound body includes a first end face and a second end face. The first end face faces the positive electrode current collector plate in the height direction. The second end face faces the negative electrode current collector plate in the

height direction. The positive electrode includes a positive electrode current collector, a positive electrode active material layer, and an insulating layer. The positive electrode active material layer covers a part of the positive electrode current collector. The positive electrode includes a positive electrode current collector covered region and a positive electrode current collector exposed region. The positive electrode current collector covered region is a region in which the positive electrode current collector is covered with the positive electrode active material layer. The positive electrode current collector exposed region is a region in which the positive electrode current collector is exposed without being covered with the positive electrode active material layer. All or a part of the positive electrode current collector exposed region forms the first end face and is coupled to the positive electrode current collector plate. The negative electrode includes a negative electrode current collector and a negative electrode active material layer. The negative electrode active material layer covers a part of the negative electrode current collector. The negative electrode includes a negative electrode current collector covered region and a negative electrode current collector exposed region. The negative electrode current collector covered region is a region in which the negative electrode current collector is covered with the negative electrode active material layer. The negative electrode current collector exposed region is a region in which the negative electrode current collector is exposed without being covered with the negative electrode active material layer. All or a part of the negative electrode current collector exposed region forms the second end face and is coupled to the negative electrode current collector plate. The insulating layer covers an opposing part, of the positive electrode current collector exposed region, that is opposed to the negative electrode active material layer with the separator interposed therebetween, and has an elongation of 180% or higher.

[0009] In the secondary battery according to an embodiment of the present disclosure, the insulating layer that covers the opposing part, of the positive electrode current collector exposed region, that is opposed to the negative electrode active material layer with the separator interposed therebetween, has the elongation of 180% or higher. Therefore, even when foreign matter such as metal powder is trapped between the insulating layer and the negative electrode active material layer, it is possible to effectively prevent the insulating layer from being broken or cracked.

[0010] The secondary battery according to an embodiment of the present disclosure thus makes it possible to achieve superior safety.

[0011] Note that effects of the present disclosure are not necessarily limited to those described above and may include any of a series of effects described below in relation to the present disclosure.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0012] FIG. **1** is a sectional diagram illustrating a configuration example of a vertical sectional structure of a secondary battery according to an embodiment of the present disclosure along a height direction.

[0013] FIG. **2** is a schematic diagram illustrating a configuration example of a stacked body including a positive electrode, a negative electrode, and a separator illustrated in FIG. **1**.

[0014] FIG. **3** is a sectional diagram illustrating a configuration example of a horizontal sectional structure of an electrode wound body illustrated in FIG. **1**.

[0015] FIG. **4**A is a developed view of the positive electrode illustrated in FIG. **1**.

[0016] FIG. **4**B is a sectional view of the positive electrode illustrated in FIG. **1**.

[0017] FIG. **5**A is a developed view of the negative electrode illustrated in FIG. **1**.

[0018] FIG. **5**B is a sectional view of the negative electrode illustrated in FIG. **1**.

[0019] FIG. **6** is an enlarged sectional diagram illustrating, in an enlarged manner, a configuration example of a part of the vertical sectional structure of the secondary battery illustrated in FIG. **1**.

[0020] FIG. **7**A is a plan view of a positive electrode current collector plate illustrated in FIG. **1**.

[0021] FIG. **7**B is a plan view of a negative electrode current collector plate illustrated in FIG. **1**.

[0022] FIG. **8** is a perspective diagram describing a process of manufacturing the secondary battery illustrated in FIG. **1**.

[0023] FIG. **9** is a block diagram illustrating a circuit configuration of a battery pack to which the secondary battery according to an embodiment of the present disclosure is applied.

DETAILED DESCRIPTION

[0024] The present disclosure is described below in further detail including with reference to the drawings according to an embodiment.

[0025] A description is given first of a secondary battery according to an embodiment of the present disclosure.

[0026] In the present embodiment, a cylindrical lithium-ion secondary battery having an outer appearance of a cylindrical shape will be described as an example. However, the secondary battery of the present disclosure is not limited to the cylindrical lithium-ion secondary battery, and may be a lithium-ion secondary battery having an outer appearance of a shape other than the cylindrical shape, or may be a battery in which an electrode reactant other than lithium is used.

[0027] Although a charge and discharge principle of the secondary battery is not particularly limited, the following description deals with a case where a battery capacity is obtained through insertion and extraction of the electrode reactant. The secondary battery includes a positive electrode, a negative electrode, and an electrolyte. In the secondary battery, to prevent precipitation of the electrode reactant on a surface of the negative electrode during charging, a charge capacity of the negative electrode is greater than a discharge capacity of the positive electrode. In other words, an electrochemical capacity per unit area of the negative electrode is set to be greater than an electrochemical capacity per unit area of the positive electrode.

[0028] Although not particularly limited in kind as described above, the electrode reactant is specifically a light metal such as an alkali metal or an alkaline earth metal. Examples of the alkali metal include lithium, sodium, and potassium. Examples of the alkaline earth metal include beryllium, magnesium, and calcium.

[0029] In the following, described as an example is a case where the electrode reactant is lithium. A secondary battery in which the battery capacity is obtained through insertion and extraction of lithium is what is called a lithium-ion secondary battery. In the lithium-ion secondary battery, lithium is inserted and extracted in an ionic state.

[0030] FIG. **1** illustrates a vertical sectional configuration of a lithium-ion secondary battery **1** (hereinafter simply referred to as a secondary battery **1**) according to the present embodiment along a height direction. The secondary battery **1** illustrated in FIG. **1** includes an outer package can **11** and an electrode wound body **20**. The outer package can **11** has a substantially cylindrical shape. The electrode wound body **20** is contained inside the outer package can **11** and serves as a battery device. The secondary battery **1** further includes an outer package tube **50**. The outer package tube **50** covers an outer peripheral surface of the outer package can **11**. Note that, in the present specification, the height direction of the secondary battery **1** corresponds to a Z-axis direction.

[0031] Specifically, the secondary battery **1** includes, inside the outer package can **11**, a pair of insulating plates **12** and **13**, the electrode wound body **20**, a positive electrode current collector plate **24**, and a negative electrode current collector plate **25**, for example. The positive electrode current collector plate **24** serves as a first electrode current collector plate. The negative electrode current collector plate **25** serves as a second electrode current collector plate. The electrode wound body **20** is a structure in which a positive electrode **21** and a negative electrode **22** are stacked with a separator **23** interposed therebetween and are wound, for example. The electrode wound body **20** is impregnated with an electrolytic solution. The electrolytic solution is a liquid electrolyte. Note that the secondary battery **1** may further include a thermosensitive resistive (PTC) device, a reinforcing member, or both inside the outer package can **11**.

[0032] The outer package can **11** contains components including, without limitation, the positive electrode current collector plate **24**, the negative electrode current collector plate **25**, and the electrode wound body **20**. The outer package can **11** includes a bottom part **11**B and a sidewall part **11**W. The bottom part **11**B also serves as a negative electrode terminal coupled to the negative electrode **22** via the negative electrode current collector plate **25**. The outer package can **11** has, for example, a hollow cylindrical structure having a lower end part and an upper end part in the Z-axis direction. The lower end part is closed, and the upper end part is open. The upper end part of the outer package can **11** is thus an open end part **11**N. The lower end part of the outer package can **11** is closed by the bottom part **11**B having a substantially circular plate shape. The sidewall part **11**W is provided between the open end part **11**N and the bottom part **11**B and surrounds the electrode wound body **20**. The sidewall part **11**W so stands in the height direction and along an outer edge of the bottom part **11**B as to surround the electrode wound body **20**. The sidewall part **11**W includes the open end part **11**N on an opposite side to the bottom part **11**B. The open end part **11**N is open to allow the electrode wound body **20** to be passed through the open end part **11**N. The outer package can **11** includes, for example, a metal material such as iron as a constituent material. Note that a surface of the outer package can **11** may be plated with, for example, a metal material such as nickel. The insulating plate **12** and the insulating plate **13** are so opposed to each other as to allow the electrode wound body **20** to be interposed therebetween in the Z-axis direction, for example. Note that in the present specification, the open end part **11**N and the vicinity thereof may be referred to as an upper part of the secondary battery **1** in the Z-axis direction, and a region where the outer package can **11** is closed and the vicinity thereof may be referred to as a lower part of the secondary battery **1** in the Z-axis direction.

[0033] The outer package tube **50** surrounds a side surface **11**WS that is an outer surface of the sidewall part **11**W of the outer package can **11**. The outer package tube **50** may, however, cover a bent part **11**P provided at the upper end part of the outer package can **11**, as illustrated in FIG. **1**. The bent part **11**P will be described later. Further, the outer package tube **50** may cover a part of a bottom surface **11**BS that is an outer surface of the bottom part **11**B of the outer package can **11**. The outer package tube **50** includes, for example, a thermally contractible insulating film that includes a material such as a polyester-based resin, a polyamide-based resin, or a thermoplastic elastomer resin.

[0034] A washer **55** is provided in a gap between the outer package tube **50** and the bent part **11**P of the outer package can **11**. The washer **55** is an insulating ring member that has an opening **55**K in a middle region in a plane orthogonal to the height direction. Disposed in the opening **55**K is a projecting part provided in a middle region of a battery cover **14**. The washer **55** may include a material such as black modified polyphenylene ether as a constituent material.

[0035] Each of the insulating plates **12** and **13** is, for example, a dish-shaped plate having a surface perpendicular to a central axis CL of the electrode wound body **20**, that is, a surface perpendicular to a Z-axis in FIG. **1**. The insulating plates **12** and **13** are so disposed as to allow the electrode wound body **20** to be interposed therebetween.

[0036] For example, a structure in which the battery cover **14** and a safety valve mechanism **30** are crimped with a gasket **15** interposed between the open end part **11**N and both the battery cover **14** and the safety valve mechanism **30**, that is, a crimped structure **11**R, is provided at the open end part **11**N of the outer package can **11**. The outer package can **11** is sealed by the battery cover **14**, with the electrode wound body **20** and other components being contained inside the outer package can **11**. The crimped structure **11**R is what is called a crimp structure, and includes the bent part **11**P serving as what is called a crimp part.

[0037] The battery cover **14** is a top cover that mainly covers the open end part **11**N of the outer package can **11** in a state where the electrode wound body **20** and other components are contained inside the outer package can **11**. The battery cover **14** is, for example, an electrical conductor that

includes a material similar to the material included in the outer package can **11**. The battery cover **14** closes the open end part **11**N of the outer package can **11** and is coupled to the positive electrode current collector plate **24**. Therefore, the battery cover **14** also serves as a positive electrode terminal coupled to the positive electrode **21** via the positive electrode current collector plate **24**. The middle region of the battery cover **14** protrudes upward, i.e., in a +Z direction, for example. As a result, a peripheral region, i.e., a region other than the middle region, of the battery cover **14** is in a state of being in contact with the safety valve mechanism **30**, for example. The battery cover **14** is a closing member that closes the open end part **11**N integrally with the safety valve mechanism **30**.
[0038] The gasket **15** is a sealing member interposed mainly between the bent part **11**P of the outer package can **11** and the battery cover **14**. The gasket **15** seals a gap between the bent part **11**P and the battery cover **14**. Note that a surface of the gasket **15** may be coated with, for example, asphalt. The gasket **15** includes any one or more of insulating materials, for example. The insulating material is not particularly limited in kind, and examples thereof include a polymer material such as polybutylene terephthalate (PBT) or polypropylene (PP). In particular, the insulating material is preferably polybutylene terephthalate. One reason for this is that this allows the gap between the bent part **11**P and the battery cover **14** to be sufficiently sealed, with the outer package can **11** and the battery cover **14** being electrically separated from each other.
[0039] The safety valve mechanism **30** is adapted to cancel the sealed state of the outer package can **11** to thereby release a pressure inside the outer package can **11**, i.e., an internal pressure of the outer package can **11** on an as-needed basis, mainly upon an increase in the internal pressure. Examples of a cause of the increase in the internal pressure of the outer package can **11** include a gas generated due to a decomposition reaction of the electrolytic solution upon charging and discharging. The internal pressure of the outer package can **11** can also increase due to heating from outside.
[0040] The electrode wound body **20** is disposed between the positive electrode current collector plate **24** and the negative electrode current collector plate **25**. The electrode wound body **20** has an upper end face **41** and a lower end face **42**. The upper end face **41** faces the positive electrode current collector plate **24** in the height direction. The lower end face **42** faces the negative electrode current collector plate **25** in the height direction. The electrode wound body **20** is a power generation device that causes charging and discharging reactions to proceed, and is contained inside the outer package can **11**. The electrode wound body **20** includes the positive electrode **21**, the negative electrode **22**, the separator **23**, and the electrolytic solution as a liquid electrolyte.
[0041] FIG. **2** is a developed view of the electrode wound body **20**, and schematically illustrates a part of a stacked body S**20** including the positive electrode **21**, the negative electrode **22**, and the separator **23**. The positive electrode **21** serves as a first electrode. The negative electrode **22** serves as a second electrode. In the stacked body S**20** that corresponds to the electrode wound body **20** in an unwound state, the positive electrode **21** and the negative electrode **22** are stacked on each other with the separator **23** interposed therebetween. The separator **23** includes, for example, two bases, that is, a first separator member **23**A and a second separator member **23**B. Accordingly, the electrode wound body **20** includes the stacked body S**20** that is four-layered. In the four-layered stacked body S**20**, the positive electrode **21**, the first separator member **23**A, the negative electrode **22**, and the second separator member **23**B are stacked in order. Each of the positive electrode **21**, the first separator member **23**A, the negative electrode **22**, and the second separator member **23**B is a substantially band-shaped member in which a W-axis direction corresponds to a transverse direction and an L-axis direction corresponds to a longitudinal direction.
[0042] As illustrated in FIG. **3**, the electrode wound body **20** is the stacked body S**20** so wound around a through hole **26** that extends along the central axis CL extending in the Z-axis direction as to form a spiral shape in a horizontal section orthogonal to the Z-axis direction. Here, the stacked body S**20** is wound in an orientation in which the W-axis direction substantially coincides with the Z-axis direction. Note that FIG. **3** illustrates a configuration example of the electrode wound body

20 along the horizontal section orthogonal to the Z-axis direction. Note that, for higher visibility, FIG. **3** omits illustration of the separator **23**. The electrode wound body **20** has an outer appearance of a substantially circular columnar shape as a whole. The positive electrode **21** and the negative electrode **22** are wound, remaining in a state of being opposed to each other with the separator **23** interposed therebetween. The electrode wound body **20** has the through hole **26** as an internal space at a center thereof. The through hole **26** is a hole into which a winding core for assembling the electrode wound body **20** and an electrode rod for welding are each to be put. The through hole **26** extends in the Z-axis direction along the central axis CL, and is provided through the electrode wound body **20**. The stacked body S**20** is thus wound around the through hole **26**.

[0043] The positive electrode **21**, the negative electrode **22**, and the separator **23** are so wound that the separator **23** is positioned in each of an outermost wind of the electrode wound body **20** and an innermost wind of the electrode wound body **20**. Further, in the outermost wind of the electrode wound body **20**, the negative electrode **22** is positioned on an outer side relative to the positive electrode **21**. In other words, as illustrated in FIG. **3**, an outermost positive electrode wind part **21**out that is positioned in an outermost wind of the positive electrode **21** included in the electrode wound body **20** is positioned on an inner side relative to an outermost negative electrode wind part **22**out that is positioned in an outermost wind of the negative electrode **22** included in the electrode wound body **20**. Here, the outermost positive electrode wind part **21**out is a part corresponding to the outermost one wind of the positive electrode **21** in the electrode wound body **20**. The outermost negative electrode wind part **22**out is a part corresponding to the outermost one wind of the negative electrode **22** in the electrode wound body **20**. In contrast, in the innermost wind of the electrode wound body **20**, the negative electrode **22** is positioned on the inner side relative to the positive electrode **21**. In other words, as illustrated in FIG. **3**, an innermost negative electrode wind part **22** in that is positioned in an innermost wind of the negative electrode **22** included in the electrode wound body **20** is positioned on the inner side relative to an innermost positive electrode wind part **21** in that is positioned in an innermost wind of the positive electrode **21** included in the electrode wound body **20**. Here, the innermost positive electrode wind part **21** in is a part corresponding to the innermost one wind of the positive electrode **21** in the electrode wound body **20**. The innermost negative electrode wind part **22** in is a part corresponding to the innermost one wind of the negative electrode **22** in the electrode wound body **20**. The number of winds of each of the positive electrode **21**, the negative electrode **22**, and the separator **23** is not particularly limited, and may be chosen as desired.

[0044] FIG. **4**A is a developed view of the positive electrode **21**, and schematically illustrates a state before being wound. FIG. **4**B illustrates a sectional configuration of the positive electrode **21**. Note that FIG. **4**B illustrates a section of the positive electrode **21** as viewed in an arrowed direction along line IVB-IVB illustrated in FIG. **4**A. The positive electrode **21** includes, for example, a positive electrode current collector **21**A, a positive electrode active material layer **21**B, and an insulating layer **21**C. The positive electrode active material layer **21**B covers a part of the positive electrode current collector **21**A. Note that FIG. **1** omits the illustration of the insulating layer **21**C. For example, the positive electrode active material layer **21**B may be provided only on one of two opposite surfaces of the positive electrode current collector **21**A, or may be provided on each of the two opposite surfaces of the positive electrode current collector **21**A. FIG. **4**B illustrates a case where the positive electrode active material layer **21**B is provided on each of the two opposite surfaces of the positive electrode current collector **21**A. More specifically, the positive electrode current collector **21**A includes an inward positive electrode current collector surface **21**A**1** facing toward a winding center side of the electrode wound body **20**, that is, facing toward the central axis CL, and an outward positive electrode current collector surface **21**A**2** facing toward a side opposite to the winding center side of the electrode wound body **20**, that is, positioned on a side opposite to the inward positive electrode current collector surface **21**A**1**. The positive electrode **21** includes, as the positive electrode active material layers **21**B, an inner winding side positive electrode active material layer **21**B**1** covering all or a part of the inward positive electrode current collector surface **21**A**1**, and an outer winding side positive electrode active material layer **21**B**2** covering all or a part of the outward positive electrode current collector surface **21**A**2**. Note that in the present specification, the inner winding side positive electrode active material layer **21**B**1** and the outer winding side positive electrode active material layer **21**B**2** may each be generically referred to as the positive electrode active material layer **21**B, without being distinguished from each other.

[0045] The positive electrode **21** includes a positive electrode current collector covered region **211** in which the positive electrode current collector **21**A is covered with the positive electrode active material layer **21**B, and a positive electrode current collector exposed region **212** in which the positive electrode current collector **21**A is exposed without being covered with the positive electrode active material layer **21**B. As illustrated in FIG. **4**A, the positive electrode current collector covered region **211** and the positive electrode current collector exposed region **212** each extend from a central axis side edge **21**E**1** of the positive electrode **21** to an outer winding side edge **21**E**2** of the positive electrode **21** along the L-axis direction, i.e., the longitudinal direction of the positive electrode **21**. Here, the L-axis direction corresponds to a winding direction of the electrode wound body **20**. In other words, in the positive electrode **21**, the positive electrode current collector **21**A is covered with the positive electrode active material layer **21**B from the central axis side edge **21**E**1** of the positive electrode **21** to the outer winding side edge **21**E**2** of the positive electrode **21** in the winding direction of the electrode wound body **20**. The positive electrode current collector covered region **211** and the positive electrode current collector exposed region **212** are adjacent to each other in the W-axis direction, i.e., the transverse direction of the positive electrode **21**. The W-axis direction substantially coincides with the central axis CL. Further, as illustrated in FIG. **3**, in the electrode wound body **20**, the central axis side edge **21**E**1** of the innermost positive electrode wind part **21** in is positioned to be inwardly retracted relative to a central axis side edge **22**E**1** of the innermost negative electrode wind part **22**in. The positive electrode **21** further has a lower edge **21**E**3** positioned on a side of a lower part of the electrode wound body **20** and extending in the L-axis direction. Note that FIGS. **4**A and **4**B each schematically illustrate the positive electrode current collector **21**A in a straightened state along the W-axis direction. In actuality, however, as illustrated in FIG. **1**, positive electrode edge parts **212**E of the positive electrode current collector exposed region **212** are bent toward the central axis CL and coupled to the positive electrode current collector plate **24**. In other words, an end part of the positive electrode current collector exposed region **212** in the W-axis direction forms the upper end face **41** and is coupled to the positive electrode current collector plate **24**, as illustrated in FIG. **1**. The upper end face **41** includes the positive electrode edge parts **212**E, of the positive electrode current collector exposed region **212**, that are bent toward the through hole **26** in a wound state.

[0046] The insulating layer **21**C is preferably provided in a region including a border between the positive electrode current collector covered region **211** and the positive electrode current collector exposed region **212** and the vicinity of the border. As with the positive electrode current collector covered region **211** and the positive electrode current collector exposed region **212**, the insulating layer **21**C also preferably extends from the central axis side edge **21**E**1** to the outer winding side edge **21**E**2** in the electrode wound body **20** along the L-axis direction. The insulating layer **21**C is preferably adhered to the first separator member **23**A, the second separator member **23**B, or both. One reason for this is that this makes it possible to prevent the positive electrode **21** and the separator **23** from becoming misaligned with each other. Note that a detailed configuration of the positive electrode **21** will be described later.

[0047] FIG. **5**A is a developed view of the negative electrode **22**, and schematically illustrates a state before being wound. FIG. **5**B illustrates a sectional configuration of the negative electrode **22**. Note that FIG. **5**B illustrates a section of the negative electrode **22** as viewed in an arrowed direction along line VB-VB illustrated in FIG. **5**A. The negative electrode **22** includes, for example, a negative electrode current collector **22**A and a negative electrode active material layer **22**B. The negative electrode active material layer **22**B covers a part of the negative electrode current collector **22**A. For example, the negative electrode active material layer **22**B may be provided only on one of two opposite surfaces of the negative electrode current collector **22**A, or may be provided on each of the two opposite surfaces of the negative electrode current collector **22**A. FIG. **5**B illustrates a case where the negative electrode active material layer **22**B is provided on each of the two opposite surfaces of the negative electrode current collector **22**A. More specifically, the negative electrode current collector **22**A includes an inward negative electrode current collector surface **22**A**1** facing toward the winding center side of the electrode wound body **20**, that is, facing toward the central axis CL, and an outward negative electrode current collector surface **22**A**2** facing toward the side opposite to the winding center side of the electrode wound body **20**, that is, positioned on a side opposite to the inward negative electrode current collector surface **22**A**1**. The negative electrode **22** includes, as the negative electrode active material layers **22**B, an inner winding side negative electrode active material layer **22**B**1** covering all or a part of the inward negative electrode current collector surface **22**A**1**, and an outer winding side negative electrode active material layer **22**B**2** covering all or a part of the outward negative electrode current collector surface **22**A**2**. Note that in the present specification, the inner winding side negative electrode active material layer **22**B**1** and the outer winding side negative electrode active material layer **22**B**2** may each be generically referred to as the negative electrode active material layer **22**B, without being distinguished from each other.

[0048] The negative electrode **22** includes a negative electrode current collector covered region **221** and a negative electrode current collector exposed region **222**. The negative electrode current collector covered region **221** is a region in which the negative electrode current collector **22**A is covered with the negative electrode active material layer **22**B. The negative electrode current collector exposed region **222** is a region in which the negative electrode current collector **22**A is exposed without being covered with the negative electrode active material layer **22**B. As illustrated in FIG. **5**A, the negative electrode current collector covered region **221** and the negative electrode current collector exposed region **222** each extend along the L-axis direction, i.e., the longitudinal direction of the negative electrode **22**. The negative electrode current collector exposed region **222** extends from the central axis side edge **22**E**1** of the negative electrode **22** to an outer winding side edge **22**E**2** of the negative electrode **22** in the winding direction of the electrode wound body **20**. In contrast, the negative electrode current collector covered region **221** is provided at neither the central axis side edge **22**E**1** of the negative electrode **22** nor the outer winding side edge **22**E**2** of the negative electrode **22**. As illustrated in FIG. **5**A, portions of the negative electrode current collector exposed region **222** are so provided as to allow the negative electrode current collector covered region **221** to be interposed therebetween in the L-axis direction, i.e., the longitudinal direction of the negative electrode **22**. Specifically, the negative electrode current collector exposed region **222** includes a first part **222**A, a second part **222**B, and a third part **222**C. The negative electrode **22** further has a lower edge **22**E**3** positioned on the side of the lower part of the electrode wound body **20** and extending in the L-axis direction. The first part **222**A is provided to be adjacent to the negative electrode current collector covered region **221** in the W-axis direction, and extends from the central axis side edge **22**E**1** of the negative electrode **22** to the outer winding side edge **22**E**2** of the negative electrode **22** in the L-axis direction. The second part **222**B and the third part **222**C are so provided as to allow the negative electrode current collector covered region **221** to be interposed therebetween in the L-axis direction. The first part **222**A is positioned in a region including the lower edge **22**E**3** of the negative electrode **22** and the vicinity of the lower edge **22**E**3**. The second part **222**B is positioned in a region including the central axis side edge **22**E**1** of the negative electrode **22** and the vicinity thereof, for example. The third part **222**C is positioned in a region including the outer winding side edge **22**E**2** of the negative electrode **22** and the vicinity thereof. Note that FIGS. **5**A and **5**B each schematically illustrate the negative electrode current

collector **22**A in a state of being straightened along the W-axis direction. In actuality, however, as illustrated in FIG. **1**, negative electrode edge parts **222**E of the negative electrode current collector exposed region **222** are bent toward the central axis CL and coupled to the negative electrode current collector plate **25**. In other words, an end part of the negative electrode current collector exposed region **222** in the W-axis direction forms the lower end face **42** and is coupled to the negative electrode current collector plate **25**, as illustrated in FIG. **1**. The lower end face **42** includes the negative electrode edge parts **222**E, of the negative electrode current collector exposed region **222**, that are bent toward the through hole **26** in a wound state. Note that, detailed configuration of the negative electrode **22** will be described later.
[0049] In the stacked body S**20** in the electrode wound body **20**, the positive electrode **21** and the negative electrode **22** are so stacked with the separator **23** interposed therebetween that the positive electrode current collector exposed region **212** and the first part **222**A of the negative electrode current collector exposed region **222** face toward mutually opposite directions along the W-axis direction, i.e., a width direction. In the electrode wound body **20**, an end part of the separator **23** is fixed by attaching a fixing tape **46** to a side surface part **45** of the electrode wound body **20**, which prevents loosening of winding.
[0050] In the secondary battery **1**, as illustrated in FIG. **2**, A>B is preferably satisfied, where A is a width of the positive electrode current collector exposed region **212**, and B is a width of the first part **222**A of the negative electrode current collector exposed region **222**. For example, when the width A is 7 (mm), the width B is 4 (mm). Further, C>D is preferably satisfied, where C is a width of a part of the positive electrode current collector exposed region **212** protruding from an outer edge in the width direction of the separator **23**, and D is a width of a part of the first part **222**A of the negative electrode current collector exposed region **222** protruding from an opposite outer edge in the width direction of the separator **23**. For example, when the width C is 4.5 (mm), the width D is 3 (mm). Note that the values of the widths A to D and a high-and-low relationship between the values of the widths A to D described here are merely examples, and the present disclosure is not limited thereto. Additionally, the length of the negative electrode active material layer **22**B (the negative electrode current collector covered region **221**) in the Z-axis direction is longer than the length of the positive electrode active material layer **21**B (the positive electrode current collector covered region **211**) in the Z-axis direction. In other words, a width W**22**B of the negative electrode active material layer **22**B is wider than a width W**21**B of the positive electrode active material layer **21**B (W**21**B<W**22**B). One reason for this is that having the entire positive electrode active material layer **21**B opposed to the negative electrode active material layer **22**B with the separator **23** interposed therebetween prevents precipitation of lithium ions extracted from the positive electrode **21** on a surface of the negative electrode **22**.
[0051] As illustrated in FIG. **1**, in the upper part of the secondary battery **1**, the multiple positive electrode edge parts **212**E, of the positive electrode current collector exposed region **212** wound around the central axis CL, that are adjacent to each other in a radial direction (an R direction) of the electrode wound body **20** are so bent toward the central axis CL as to overlap each other to thereby form the upper end face **41** of the electrode wound body **20**. Similarly, in the lower part of the secondary battery **1**, the multiple negative electrode edge parts **222**E, of the negative electrode current collector exposed region **222** wound around the central axis CL, that are adjacent to each other in the radial direction (the R direction) are so bent toward the central axis CL as to overlap each other to thereby form the lower end face **42** of the electrode wound body **20**. Accordingly, the multiple positive electrode edge parts **212**E of the positive electrode current collector exposed region **212** gather at the upper end face **41** of the electrode wound body **20**, and the multiple negative electrode edge parts **222**E of the negative electrode current collector exposed region **222** gather at the lower end face **42** of the electrode wound body **20**. To achieve better contact between the positive electrode current collector plate **24** that is provided for extracting a current and the positive electrode edge parts **212**E, the multiple positive electrode edge parts **212**E are bent toward

the central axis CL and form a flat surface. Similarly, to achieve better contact between the negative electrode current collector plate **25** that is provided for extracting a current and the negative electrode edge parts **222**E, the multiple negative electrode edge parts **222**E are bent toward the central axis CL and form a flat surface. Note that as used herein, the term “flat surface” encompasses not only a completely flat surface but also a surface having some asperities or surface roughness to the extent that joining of the positive electrode current collector exposed region **212** to the positive electrode current collector plate **24** and joining of the negative electrode current collector exposed region **222** to the negative electrode current collector plate **25** are possible.

[0052] The positive electrode current collector **21**A includes, for example, an aluminum foil, as will be described later. In contrast, the negative electrode current collector **22**A includes, for example, a copper foil, as will be described later. In this case, the positive electrode current collector **21**A is softer than the negative electrode current collector **22**A. In other words, the positive electrode current collector exposed region **212** has a Young's modulus lower than a Young's modulus of the negative electrode current collector exposed region **222**. Accordingly, in an embodiment, it is more preferable that the widths A to D satisfy a relationship of A>B and C>D. In such a case, when the positive electrode current collector exposed region **212** and the negative electrode current collector exposed region **222** are simultaneously bent with equal pressures from respective electrode sides, the bent portion in the positive electrode **21** and the bent portion in the negative electrode **22** may sometimes have substantially equal heights measured from respective ends of the separator **23**. In this case, the multiple positive electrode edge parts **212**E (FIG. **1**) of the positive electrode current collector exposed region **212** appropriately overlap each other by being bent. This allows for easy joining of the positive electrode current collector exposed region **212** and the positive electrode current collector plate **24** to each other. Similarly, the multiple negative electrode edge parts **222**E (FIG. **1**) of the negative electrode current collector exposed region **222** appropriately overlap each other by being bent. This allows for easy joining of the negative electrode current collector exposed region **222** and the negative electrode current collector plate **25** to each other. As used herein, the term “joining” refers to coupling by, for example, laser welding; however, a method of joining is not limited to the laser welding.

[0053] FIG. **6** is an enlarged sectional diagram illustrating, in an enlarged manner, a configuration example of a part of a vertical sectional structure of the secondary battery **1** illustrated in FIG. **1**. As illustrated in FIGS. **2** and **6**, an opposing part FP, of the positive electrode current collector exposed region **212** of the positive electrode **21**, that is opposed to the negative electrode **22** with the separator **23** interposed therebetween is covered with the insulating layer **21**C. The insulating layer **21**C has a width of 3 mm in the W-axis direction, for example. The insulating layer **21**C entirely covers a part of the positive electrode current collector exposed region **212** of the positive electrode **21**, that is opposed to the negative electrode current collector covered region **221** of the negative electrode **22** with the separator **23** interposed therebetween. The insulating layer **21**C makes it possible to effectively prevent an internal short circuit of the secondary battery **1** when foreign matter enters between the negative electrode current collector covered region **221** and the positive electrode current collector exposed region **212**, for example. Further, when the secondary battery **1** undergoes an impact, the insulating layer **21**C absorbs the impact and thereby makes it possible to effectively prevent bending of the positive electrode current collector exposed region **212** and a short circuit between the positive electrode current collector exposed region **212** and the negative electrode **22**.

[0054] The insulating layer **21**C covers the opposing part FP (to be described later), of the positive electrode current collector exposed region **212**, that is opposed to the negative electrode active material layer **22**B (to be described later) with the separator **23** interposed therebetween. The insulating layer **21**C has an elongation of 180% or higher, for example. Further, the insulating layer **21**C preferably includes a resin including modified polyvinylidene difluoride (PVDF) or copolymer PVDF. One reason for this is that when the insulating layer **21**C includes modified PVDF or copolymer PVDF, the insulating layer **21**C is swollen by, for example, a solvent included in the electrolytic solution, which allows the insulating layer **21**C to be favorably adhered to the separator **23**. Another reason for this is that when the insulating layer **21**C includes modified PVDF or copolymer PVDF, a more favorable elongation is obtained. Examples of the constituent material included in the insulating layer **21**C include PVDF being modified or copolymerized with tetrafluoroethylene (TFE), hexafluoropropylene (HFP), acrylic acid, or maleic acid.

[0055] The secondary battery **1** may further include insulating tapes **53** and **54** in a gap between the outer package can **11** and the electrode wound body **20**. The positive electrode current collector exposed region **212** having portions gathering at the upper end face **41** and the negative electrode current collector exposed region **222** having portions gathering at the lower end face **42** include electrical conductors, such as metal foils, that are exposed. Accordingly, if the positive electrode current collector exposed region **212** and the negative electrode current collector exposed region **222** are in proximity to the outer package can **11**, a short circuit between the positive electrode **21** and the negative electrode **22** can occur via the outer package can **11**. A short circuit can also occur when the positive electrode current collector plate **24** on the upper end face **41** and the outer package can **11** come into close proximity to each other. To address this, it is preferable to provide the insulating tapes **53** and **54** as insulating members. Each of the insulating tapes **53** and **54** is an adhesive tape including a base layer, and an adhesive layer provided on one surface of the base layer. The base layer includes, for example, any of polypropylene, polyethylene terephthalate, or polyimide. To prevent the provision of the insulating tapes **53** and **54** from resulting in a decreased capacity of the electrode wound body **20**, the insulating tapes **53** and **54** are disposed not to overlap the fixing tape **46** attached to the side surface part **45**, and each have a thickness set to be less than or equal to a thickness of the fixing tape **46**.

[0056] In a typical lithium-ion secondary battery, for example, a lead for current extraction is welded to one location on each of the positive electrode and the negative electrode. However, such a structure increases the internal resistance of the lithium-ion secondary battery and causes the lithium-ion secondary battery to generate heat and become hot upon discharging; therefore, the structure is unsuitable for high-rate discharging. To address this, in the secondary battery **1** according to the present embodiment, the positive electrode current collector plate **24** is disposed to face the upper end face **41**, and the negative electrode current collector plate **25** is disposed to face the lower end face **42**. In addition, the positive electrode current collector covered region **211** present at the upper end face **41** and the positive electrode current collector plate **24** are welded to each other at multiple points; and the negative electrode current collector covered region **221** present at the lower end face **42** and the negative electrode current collector plate **25** are welded to each other at multiple points. A reduced internal resistance of the secondary battery **1** is thereby achieved. Each of the upper end face **41** and the lower end face **42** being a flat surface as described above also contributes to the reduced resistance. The positive electrode current collector plate **24** is disposed between the battery cover **14** and the upper end face **41**. The positive electrode current collector plate **24** is electrically coupled to the battery cover **14** via the safety valve mechanism **30**, for example. The negative electrode current collector plate **25** is disposed between the bottom part **11**B of the outer package can **11** and the lower end face **42**. The negative electrode current collector plate **25** is electrically coupled to an inner surface of the bottom part **11**B of the outer package can **11**, for example. FIG. **7**A is a developed diagram illustrating a configuration example of the positive electrode current collector plate **24**. FIG. **7**B is a developed diagram illustrating a configuration example of the negative electrode current collector plate **25**. The positive electrode current collector plate **24** is a metal plate including, for example, aluminum or an aluminum alloy as a single component, or a composite material of aluminum and the aluminum alloy. The negative electrode current collector plate **25** is a metal plate including, for example, nickel, a nickel alloy, copper, or a copper alloy as a single component, or a composite material of two or more thereof.

[0057] As illustrated in FIG. **7**A, the positive electrode current collector plate **24** includes a fan-shaped part **31** and a band-shaped part **32**. The fan-shaped part **31** has a substantially fan shape. The band-shaped part **32** has a substantially rectangular shape. A shape of the positive electrode current collector plate **24** is, however, not limited to the shape illustrated in FIG. **7**A, and may be chosen as desired. Note that in the secondary battery **1**, the positive electrode current collector plate **24** is contained inside the outer package can **11**, as illustrated in FIG. **1**, in a state where the band-shaped part **32** is bent with respect to the fan-shaped part **31**. FIG. **7**A illustrates the positive electrode current collector plate **24** in an unbent state. The fan-shaped part **31** is a facing part facing and coupled to the upper end face **41**. The fan-shaped part **31** has an outer edge including a linear part and a curved part, for example. The fan-shaped part **31** has an opening **35** in the vicinity of a middle thereof. FIG. **7**A illustrates an example case where the opening **35** has a circular plan shape in a horizontal plane orthogonal to the Z-axis direction. The band-shaped part **32** is coupled to the linear part of the outer edge of the fan-shaped part **31**, for example. The band-shaped part **32** extends in a direction intersecting with the linear part of the fan-shaped part **31**. As illustrated in FIG. **1**, in the secondary battery **1**, the positive electrode current collector plate **24** is so provided as to allow the opening **35** to overlap the through hole **26** in the Z-axis direction. In other words, the opening **35** is positioned to overlap, in the Z-axis direction, a part of the upper end face **41** on the winding center side.

[0058] A hatched portion in FIG. **7**A represents an insulating part **32**A of the band-shaped part **32**. The insulating part **32**A is a part of the band-shaped part **32** and has an insulating tape attached thereto or an insulating material applied thereto. Of the band-shaped part **32**, a portion below the insulating part **32**A is a coupling part **32**B to be coupled to a sealing plate that also serves as an external terminal. The sealing plate is electrically continuous with the battery cover **14**. Note that when the secondary battery **1** has a battery structure without a metallic center pin in the through hole **26** as illustrated in FIG. **1**, there is a low possibility that the band-shaped part **32** will come into contact with a region of a negative electrode potential. In such a case, the positive electrode current collector plate **24** does not have to include the insulating part **32**A. When the positive electrode current collector plate **24** does not include the insulating part **32**A, it is possible to increase a width of each of the positive electrode **21** and the negative electrode **22** by an amount corresponding to a thickness of the insulating part **32**A to thereby increase a charge and discharge capacity.

[0059] The negative electrode current collector plate **25** illustrated in FIG. **7**B has a shape similar to the shape of the positive electrode current collector plate **24** illustrated in FIG. **7**A. The negative electrode current collector plate **25** includes a fan-shaped part **33** and a band-shaped part **34**. The fan-shaped part **33** has a substantially fan shape. The band-shaped part **34** has a substantially rectangular shape. The shape of the negative electrode current collector plate **25** is, however, not limited to the shape illustrated in FIG. **7**B, and may be chosen as desired. Note that in the secondary battery **1**, the negative electrode current collector plate **25** is contained inside the outer package can **11**, as illustrated in FIG. **1**, in a state where the band-shaped part **34** is bent with respect to the fan-shaped part **33**. FIG. **7**B illustrates the negative electrode current collector plate **25** in an unbent state. The fan-shaped part **33** is a facing part facing and coupled to the lower end face **42**. The fan-shaped part **33** has an outer edge including a linear part and a curved part, for example. The band-shaped part **34** is coupled to the linear part of the outer edge of the fan-shaped part **33**, for example. The band-shaped part **34** extends in a direction intersecting with the linear part of the fan-shaped part **33**. The band-shaped part **34** of the negative electrode current collector plate **25** is shorter than the band-shaped part **32** of the positive electrode current collector plate **24**, and includes no portion corresponding to the insulating part **32**A of the positive electrode current collector plate **24**. The band-shaped part **34** is provided with projections **37** that are depicted as circles. The projections **37** each have a round shape. All or a part of the projections **37** is welded to the bottom part **11**B of the outer package can **11**. Upon resistance welding, a current is concentrated on the projections **37**, causing the projections **37** to melt to cause the band-shaped part **34** to be welded to the bottom part **11**B of the outer package can **11**. As with the positive electrode current collector plate **24**, the negative electrode current collector plate **25** has an opening **36** in the vicinity of a middle of the fan-shaped part **33**. In the secondary battery **1**, the negative electrode current collector plate **25** is so provided as to allow the opening **36** to overlap the through hole **26** in the Z-axis direction. FIG. **7**B illustrates an example case where the opening **36** has a circular plan shape in a horizontal plane orthogonal to the Z-axis direction.

[0060] The fan-shaped part **31** of the positive electrode current collector plate **24** simply covers a part of the upper end face **41**, owing to a plan shape of the fan-shaped part **31**. Similarly, the fan-shaped part **33** of the negative electrode current collector plate **25** simply covers a part of the lower end face **42**, owing to a plan shape of the fan-shaped part **33**. Reasons why the fan-shaped part **31** and the fan-shaped part **33** do not respectively cover the entire upper end face **41** and the entire lower end face **42** include the following two reasons, for example. A first reason is to allow the electrolytic solution to smoothly permeate the electrode wound body **20** in assembling the secondary battery **1**, for example. In particular, in the secondary battery **1** according to the present embodiment, the positive electrode current collector plate **24** is so provided as to allow the opening **35** to overlap a part of the upper end face **41** on the winding center side in the Z-axis direction. Accordingly, portions of the positive electrode edge parts **212**E forming the upper end face **41** are not covered with the fan-shaped part **31** of the positive electrode current collector plate **24** and are exposed from the opening **35**. The secondary battery **1** thus has a structure that allows for swifter permeation of the electrolytic solution into the electrode wound body **20**. A second reason is to allow a gas generated when the lithium-ion secondary battery comes into an abnormally hot state or an overcharged state to be easily released to the outside.

[0061] The positive electrode current collector **21**A includes an electrically conductive material such as aluminum, for example. The positive electrode current collector **21**A is a metal foil including aluminum or an aluminum alloy, for example.

[0062] The positive electrode active material layer **21**B includes, as a positive electrode active material, any one or more of positive electrode materials into which lithium is insertable and from which lithium is extractable. Note that the positive electrode active material layer **21**B may further include any one or more of other materials. Examples of the other materials include a positive electrode binder and a positive electrode conductor. It is preferable that the positive electrode material be a lithium-containing compound, and more specifically, a lithium-containing composite oxide or a lithium-containing phosphoric acid compound, for example. The lithium-containing composite oxide is an oxide including lithium and one or more of other elements, that is, one or more of elements other than lithium, as constituent elements. The lithium-containing composite oxide has any of crystal structures including, without limitation, a layered rock-salt crystal structure and a spinel crystal structure, for example. The lithium-containing phosphoric acid compound is a phosphoric acid compound including lithium and one or more of other elements as constituent elements, and has a crystal structure such as an olivine crystal structure, for example. The positive electrode active material layer **21**B preferably includes, as the positive electrode active material, at least one of lithium cobalt oxide, lithium nickel cobalt manganese oxide, or lithium nickel cobalt aluminum oxide, in particular. The positive electrode binder includes, for example, any one or more of materials including, without limitation, a synthetic rubber and a polymer compound. Examples of the synthetic rubber include a styrene-butadiene-based rubber, a fluorine-based rubber, and ethylene propylene diene. Examples of the polymer compound include polyvinylidene difluoride and polyimide. The positive electrode conductor includes, for example, any one or more of materials including, without limitation, a carbon material. Examples of the carbon material include graphite, carbon black, acetylene black, and Ketjen black. Note that the positive electrode conductor may be any of electrically conductive materials, and may be, for example, a metal material or an electrically conductive polymer.

[0063] The negative electrode current collector **22**A includes an electrically conductive material such as copper, for example. The negative electrode current collector **22**A is a metal foil including, for example, nickel, a nickel alloy, copper, or a copper alloy. A surface of the negative electrode current collector **22**A is preferably roughened. One reason for this is that this improves adherence of the negative electrode active material layer **22**B to the negative electrode current collector **22**A owing to what is called an anchor effect. In this case, it may suffice that the surface of the negative electrode current collector **22**A is roughened at least in a region facing the negative electrode active material layer **22**B. Examples of a roughening method include a method in which microparticles are formed through an electrolytic treatment. In the electrolytic treatment, the microparticles are formed on the surface of the negative electrode current collector **22**A by an electrolytic method in an electrolyzer. This provides the surface of the negative electrode current collector **22**A with asperities. A copper foil produced by the electrolytic method is generally called an electrolytic copper foil.

[0064] The negative electrode active material layer **22**B includes, as a negative electrode active material, any one or more of negative electrode materials into which lithium is insertable and from which lithium is extractable. Note that the negative electrode active material layer **22**B may further include any one or more of other materials. Examples of the other materials include a negative electrode binder and a negative electrode conductor. For example, the negative electrode material is a carbon material. One reason for this is that the carbon material exhibits very little change in crystal structure at the time of insertion and extraction of lithium, and a high energy density is thus obtainable stably. Another reason is that the carbon material also serves as a negative electrode conductor, which allows the negative electrode active material layer **22**B to be improved in electrically conductive property. Examples of the carbon material include graphitizable carbon, non-graphitizable carbon, and graphite. Note that spacing of a (002) plane of the non-graphitizable carbon is preferably 0.37 nm or greater. Spacing of a (002) plane of the graphite is preferably 0.34 nm or less. More specific examples of the carbon material include pyrolytic carbons, cokes, glassy carbon fibers, an organic polymer compound fired body, activated carbon, and carbon blacks. Examples of the cokes include pitch coke, needle coke, and petroleum coke. The organic polymer compound fired body is a resultant of firing or carbonizing a polymer compound such as a phenol resin or a furan resin at a suitable temperature. Other than the above, the carbon material may be low-crystalline carbon heat-treated at a temperature of about 1000° C. or lower, or may be amorphous carbon. Note that the carbon material may have any of a fibrous shape, a spherical shape, a granular shape, and a flaky shape. In the secondary battery **1**, when an open circuit voltage in a fully charged state, that is, a battery voltage is 4.25 V or higher, the amount of extracted lithium per unit mass increases as compared with when the open circuit voltage in the fully charged state is 4.20 V, even with the same positive electrode active material. The amount of the positive electrode active material and the amount of the negative electrode active material are therefore adjusted accordingly. This makes it possible to obtain a high energy density.

[0065] The negative electrode active material layer **22**B may include, as the negative electrode active material, a silicon-containing material including at least one of silicon, silicon oxide, a carbon-silicon compound, or a silicon alloy. The term “silicon-containing material” is a generic term for a material that includes silicon as a constituent element. Note that the silicon-containing material may include only silicon as the constituent element. One silicon-containing material may be used, or two or more silicon-containing materials may be used. The silicon-containing material is able to form an alloy with lithium, and may be a simple substance of silicon, a silicon alloy, a silicon compound, a mixture of two or more thereof, or a material including one or more phases thereof. Further, the silicon-containing material may be crystalline or amorphous, or may include both a crystalline part and an amorphous part. Note that the simple substance described here refers to a simple substance merely in a general sense. The simple substance may thus include a small amount of impurity. In other words, purity of the simple substance is not limited to 100%. The silicon alloy includes, as one or more constituent elements other than silicon, any one or more of elements including, without limitation, tin, nickel, copper, iron, cobalt, manganese, zinc, indium, silver, titanium, germanium, bismuth, antimony, and chromium, for example. The silicon compound includes, as one or more constituent elements other than silicon, any one or more of elements including, without limitation, carbon and oxygen, for example. Note that the silicon compound may include, as one or more constituent elements other than silicon, any one or more of the series of constituent elements described above in relation to the silicon alloy, for example. Specific examples of the silicon alloy and the silicon compound include SiB_4, SiB_6, Mg_2Si, Ni_2Si, $TiSi_2$, $MoSi_2$, $CoSi_2$, $NiSi_2$, $CaSi_2$, $CrSi_2$, Cu_5Si, $FeSi_2$, $MnSi_2$, $NbSi_2$, $TaSi_2$, VSi_2, WSi_2, $ZnSi_2$, SiC, Si_3N_4, Si_2N_2O, and SiO_v (where $0<v\leq2$). Note that v may be set within any desired range, and may, for example, fall within the following range: $0.2<v<1.4$.

[0066] The separator **23** is interposed between the positive electrode **21** and the negative electrode **22**. The separator **23** allows lithium ions to pass through and prevents a short circuit of a current caused by contact between the positive electrode **21** and the negative electrode **22**. The separator **23** includes, for example, any one or more kinds of porous films each including, for example, a synthetic resin or a ceramic, and may include a stacked film of two or more kinds of porous films. Examples of the synthetic resin include polytetrafluoroethylene, polypropylene, and polyethylene. Note that the separator **23** preferably includes the bases that each include a single-layer polyolefin porous film including polyethylene. One reason for this is that a favorable high output characteristic is obtainable as compared with a stacked film. When the first separator member **23**A and the second separator member **23**B included in the separator **23** each include a single-layer porous film including polyolefin, the porous film preferably has a thickness of greater than or equal to 10 μm and less than or equal to 15 μm, for example. An internal short circuit is sufficiently avoidable if the single-layer porous film including polyolefin has a thickness of greater than or equal to 10 μm. A more favorable discharge capacity characteristic is achievable if the thickness of the single-layer porous film including polyolefin is less than or equal to 15 μm. Further, the porous film preferably has a surface density of greater than or equal to 6.3 g/m^2 and less than or equal to 8.3 g/m^2, for example. Allowing the single-layer porous film including polyolefin to have a surface density of greater than or equal to 6.3 g/m^2 makes it possible to sufficiently avoid an internal short circuit. Allowing the single-layer porous film including polyolefin to have a surface density of less than or equal to 8.3 g/m^2 achieves a more favorable discharge capacity characteristic.

[0067] In particular, the separator **23** may include, for example, the porous film as each of the above-described bases, and a polymer compound layer provided on one of or each of two opposite surfaces of each of the bases. One reason for this is that adherence of the separator **23** to each of the positive electrode **21** and the negative electrode **22** improves, which suppresses distortion of the electrode wound body **20**. As a result, a decomposition reaction of the electrolytic solution is suppressed, and leakage of the electrolytic solution with which the bases are impregnated is also suppressed. This prevents an easy increase in resistance even upon repeated charging and discharging, and also suppresses swelling of the battery. The polymer compound layer includes a polymer compound such as polyvinylidene difluoride, for example. One reason for this is that the polymer compound such as polyvinylidene difluoride has superior physical strength and is electrochemically stable. Note that the polymer compound may be other than polyvinylidene difluoride. To form the polymer compound layer, for example, a solution in which the polymer compound is dissolved in a solvent such as an organic solvent is applied on the base, following which the base is dried. Alternatively, the base may be immersed in the solution and thereafter dried. The polymer compound layer may include any one or more kinds of insulating particles such as inorganic particles, for example. Examples of the kind of the inorganic particles include aluminum oxide and aluminum nitride.

[0068] The electrolytic solution includes a solvent and an electrolyte salt. Note that the electrolytic solution may further include any one or more of other materials. Examples of the other materials include an additive. The solvent includes any one or more of nonaqueous solvents including, without limitation, an organic solvent. An electrolytic solution including a nonaqueous solvent is what is called a nonaqueous electrolytic solution. The nonaqueous solvent includes a fluorine compound and a dinitrile compound, for example. The fluorine compound includes, for example, at least one of fluorinated ethylene carbonate, trifluorocarbonate, trifluoroethyl methyl carbonate, a fluorinated carboxylic acid ester, or a fluorine ether. The nonaqueous solvent may further include at least one of nitrile compounds other than the dinitrile compound. Examples of the nitrile compounds other than the dinitrile compound include a mononitrile compound and a trinitrile compound. For example, succinonitrile (SN) is preferable as the dinitrile compound. Note that the dinitrile compound is not limited to succinonitrile, and may be another dinitrile compound such as adiponitrile, for example.

[0069] The electrolyte salt includes, for example, any one or more of salts including, without limitation, a lithium salt. Note that the electrolyte salt may include a salt other than the lithium salt, for example. Examples of the salt other than lithium salt include a salt of a light metal other than lithium. Examples of the lithium salt include lithium hexafluorophosphate (LiPF.sub.6), lithium tetrafluoroborate (LiBF.sub.4), lithium perchlorate (LiClO.sub.4), lithium hexafluoroarsenate (LiAsF.sub.6), lithium tetraphenylborate (LiB(C.sub.6H.sub.5).sub.4), lithium methanesulfonate (LiCH.sub.3SO.sub.3), lithium trifluoromethanesulfonate (LiCF.sub.3SO.sub.3), lithium tetrachloroaluminate (LiAlCl.sub.4), dilithium hexafluorosilicate (Li.sub.2SiF.sub.6), lithium chloride (LiCl), and lithium bromide (LiBr). In particular, the lithium salt is preferably any one or more of lithium hexafluorophosphate, lithium tetrafluoroborate, lithium perchlorate, or lithium hexafluoroarsenate, and more preferably, lithium hexafluorophosphate. Although not particularly limited, a content of the electrolyte salt is preferably within a range from 0.3 mol/kg to 3 mol/kg both inclusive with respect to the solvent, in particular. When the electrolytic solution includes LiPF.sub.6 as the electrolyte salt, a concentration of LiPF.sub.6 in the electrolytic solution is preferably higher than or equal to 1.25 mol/kg and lower than or equal to 1.45 mol/kg. One reason for this is that this makes it possible to prevent cycle deterioration caused by consumption, or decomposition, of the salt at the time of high load rate charging, and thus allows for improvement in high-load cyclability characteristic. When the electrolytic solution further includes LiBF.sub.4 in addition to LiPF.sub.6 as the electrolyte salt, a concentration of LiBF.sub.4 in the electrolytic solution is preferably higher than or equal to 0.001 (wt %) and lower than or equal to 0.1 (wt %). One reason for this is that this makes it possible to more effectively prevent the cycle deterioration caused by consumption, or decomposition, of the salt at the time of high load rate charging, and thus allows for even further improvement in high-load cyclability characteristic.

[0070] In the secondary battery **1** according to the present embodiment, for example, upon charging, lithium ions are extracted from the positive electrode **21**, and the extracted lithium ions are inserted into the negative electrode **22** via the electrolytic solution. In the secondary battery **1**, for example, upon discharging, lithium ions are extracted from the negative electrode **22**, and the extracted lithium ions are inserted into the positive electrode **21** via the electrolytic solution.

[0071] A method of manufacturing the secondary battery **1** will be described with reference to FIG. **8** as well as FIGS. **1** to **7**(B). FIG. **8** is a perspective diagram describing a process of manufacturing the secondary battery illustrated in FIG. **1**.

[0072] First, the positive electrode current collector **21**A is prepared, and the positive electrode active material layer **21**B is selectively formed on the surface of the positive electrode current collector **21**A to thereby form the positive electrode **21** including the positive electrode current collector covered region **211** and the positive electrode current collector exposed region **212**. Thereafter, the negative electrode current collector **22**A is prepared, and the negative electrode active material layer **22**B is selectively formed on the surface of the negative electrode current collector **22**A to thereby form the negative electrode **22** including the negative electrode current

collector covered region **221** and the negative electrode current collector exposed region **222**. The positive electrode **21** and the negative electrode **22** may be subjected to a drying process. Thereafter, the positive electrode **21** and the negative electrode **22** are stacked, with the first separator member **23**A and the second separator member **23**B on the positive electrode **21** and the negative electrode **22**, respectively, to cause the positive electrode current collector exposed region **212** and the first part **222**A of the negative electrode current collector exposed region **222** to be on opposite sides to each other in the W-axis direction. The stacked body S**20** is thereby fabricated. Thereafter, the stacked body S**20** is so wound in a spiral shape as to form the through hole **26**. Upon thus winding the stacked body S**20**, for example, a circular columnar winding core is used as a jig, and the stacked body S**20** is wound around the circular columnar winding core. In addition, the fixing tape **46** is attached to an outermost wind of the stacked body S**20** wound in the spiral shape, following which the winding core is removed. The electrode wound body **20** is thus obtained as illustrated in part (A) of FIG. **8**.
[0073] Thereafter, as illustrated in part (B) of FIG. **8**, a part of the upper end face **41** and a part of the lower end face **42** of the electrode wound body **20** are each locally bent by pressing an end of, for example, a 0.5-millimeter-thick flat plate against each of the upper end face **41** and the lower end face **42** perpendicularly, that is, in the Z-axis direction. As a result, grooves **43** are formed to extend radiately in the radial directions (the R directions) from the through hole **26**. Note that the number and arrangement of the grooves **43** illustrated in part (B) of FIG. **8** are merely an example, and the present disclosure is not limited thereto.
[0074] Thereafter, as illustrated in part (C) of FIG. **8**, substantially equal pressures are applied to the upper end face **41** and the lower end face **42** in substantially perpendicular directions from above and below the electrode wound body **20** at substantially the same time. At this time, for example, a rod-shaped jig is placed in the through hole **26** in advance. By this operation, the positive electrode current collector exposed region **212** and the first part **222**A of the negative electrode current collector exposed region **222** are bent to thereby make each of the upper end face **41** and the lower end face **42** into a flat surface. At this time, the portions, of the positive electrode edge parts **212**E of the positive electrode current collector exposed region **212** at the upper end face **41**, that are adjacent to each other in the radial direction of the electrode wound body **20** are preferably so bent toward the through hole **26** as to overlap each other. Similarly, the portions, of the negative electrode edge parts **222**E of the negative electrode current collector exposed region **222** at the lower end face **42**, that are adjacent to each other in the radial direction of the electrode wound body **20** are preferably so bent toward the through hole **26** as to overlap each other. Thereafter, the fan-shaped part **31** of the positive electrode current collector plate **24** is joined to the upper end face **41** by a method such as the laser welding, and the fan-shaped part **33** of the negative electrode current collector plate **25** is joined to the lower end face **42** by a method such as the laser welding.
[0075] Thereafter, the insulating tapes **53** and **54** are attached to predetermined locations on the electrode wound body **20**. Thereafter, as illustrated in part (D) of FIG. **8**, the band-shaped part **32** of the positive electrode current collector plate **24** is bent and inserted through a hole **12**H of the insulating plate **12**. Further, the band-shaped part **34** of the negative electrode current collector plate **25** is bent and inserted through a hole **13**H of the insulating plate **13**.
[0076] Thereafter, the electrode wound body **20** having been assembled in the above-described manner is placed into the outer package can **11** illustrated in part (E) of FIG. **8**, following which the bottom part **11**B of the outer package can **11** and the negative electrode current collector plate **25** are welded to each other. Thereafter, a narrow part **11**S is formed in the vicinity of the open end part **11**N of the outer package can **11**. Further, the electrolytic solution is injected into the outer package can **11**, following which the band-shaped part **32** of the positive electrode current collector plate **24** and the safety valve mechanism **30** are welded to each other.
[0077] Thereafter, as illustrated in part (F) of FIG. **8**, the outer package can **11** is sealed with the

gasket **15**, the safety valve mechanism **30**, and the battery cover **14**, through the use of the narrow part **11**S. Lastly, the outer package can **11** with the washer **55** attached on the battery cover **14** is covered with the outer package tube **50**, following which the outer package tube **50** is heated by, for example, applying hot air to the outer package tube **50**. The outer package tube **50** is thus contracted and closely attached to the outer surface of the outer package can **11**.

[0078] The secondary battery **1** according to the present embodiment is completed in the above-described manner.

[0079] As described above, in the secondary battery **1** according to the present embodiment, the insulating layer **21**C that covers the opposing part FP, of the positive electrode current collector exposed region **212**, that is opposed to the negative electrode active material layer **22**B with the separator **23** interposed therebetween has the elongation of 180% or higher. Therefore, even when foreign matter such as metal powder is trapped between the insulating layer **21**C and the negative electrode active material layer **22**B, and the cell is pressurized, it is possible to effectively prevent the insulating layer **21**C from being broken or cracked owing to the extension of the insulating layer **21**C to which the foreign matter such as metal powder is pressed. This makes it possible to effectively prevent a short circuit between the positive electrode **21** and the negative electrode **22** via the foreign matter. Accordingly, it is possible for the secondary battery **1** according to the present embodiment to achieve superior safety.

[0080] In the secondary battery **1**, in particular, the insulating layer **21**C includes modified PVDF (polyvinylidene difluoride) or copolymer PVDF, resulting in achieving a more favorable elongation. This makes it possible to ensure superior safety.

[0081] Examples of applications of the secondary battery **1** according to the foregoing embodiment of the present disclosure are as described below.

[0082] FIG. **9** is a block diagram illustrating a circuit configuration example in which a battery according to an embodiment of the disclosure, which will hereinafter be referred to as a secondary battery as appropriate, is applied to a battery pack **300**. The battery pack **300** includes an assembled battery **301**, an outer package body **305**, a switcher **304**, a current detection resistor **307**, a temperature detection device **308**, and a controller **310**. The outer package body **305** contains the assembled battery **301**. The switcher **304** includes a charge control switch **302***a* and a discharge control switch **303***a*.

[0083] The battery pack **300** includes a positive electrode terminal **321** and a negative electrode terminal **322**. Upon charging, the positive electrode terminal **321** and the negative electrode terminal **322** are respectively coupled to a positive electrode terminal and a negative electrode terminal of a charger to thereby perform charging. Upon use of electronic equipment, the positive electrode terminal **321** and the negative electrode terminal **322** are respectively coupled to a positive electrode terminal and a negative electrode terminal of the electronic equipment to thereby perform discharging.

[0084] The assembled battery **301** includes multiple secondary batteries **301***a* coupled in series or in parallel. The secondary battery **1** described above is applicable to each of the secondary batteries **301***a*. Note that FIG. **9** illustrates an example case in which six secondary batteries **301***a* are coupled in a two parallel coupling and three series coupling (2P3S) configuration; however, the secondary batteries **301***a* may be coupled in any other manner such as in any n parallel coupling and m series coupling configuration (where n and m are each an integer).

[0085] The switcher **304** includes the charge control switch **302***a*, a diode **302***b*, the discharge control switch **303***a*, and a diode **303***b*, and is controlled by the controller **310**. The diode **302***b* has a polarity that is in a reverse direction with respect to a charge current flowing in a direction from the positive electrode terminal **321** to the assembled battery **301** and that is in a forward direction with respect to a discharge current flowing in a direction from the negative electrode terminal **322** to the assembled battery **301**. The diode **303***b* has a polarity that is in the forward direction with respect to the charge current and in the reverse direction with respect to the discharge current. Note that although the switcher **304** is provided on a positive side in FIG. **9**, the switcher **304** may be provided on a negative side.

[0086] The charge control switch **302***a* is so controlled by a charge and discharge controller that when the battery voltage reaches an overcharge detection voltage, the charge control switch **302***a* is turned off to thereby prevent the charge current from flowing through a current path of the assembled battery **301**. After the charge control switch **302***a* is turned off, only discharging is enabled through the diode **302***b*. Further, the charge control switch **302***a* is so controlled by the controller **310** that when a large current flows upon charging, the charge control switch **302***a* is turned off to thereby block the charge current flowing through the current path of the assembled battery **301**. The discharge control switch **303***a* is so controlled by the controller **310** that when the battery voltage reaches an overdischarge detection voltage, the discharge control switch **303***a* is turned off to thereby prevent the discharge current from flowing through the current path of the assembled battery **301**. After the discharge control switch **303***a* is turned off, only charging is enabled through the diode **303***b*. Further, the discharge control switch **303***a* is so controlled by the controller **310** that when a large current flows upon discharging, the discharge control switch **303***a* is turned off to thereby block the discharge current flowing through the current path of the assembled battery **301**.

[0087] The temperature detection device **308** is, for example, a thermistor. The temperature detection device **308** is provided in the vicinity of the assembled battery **301**, measures a temperature of the assembled battery **301**, and supplies the measured temperature to the controller **310**. A voltage detector **311** measures a voltage of the assembled battery **301** and a voltage of each of the secondary batteries **301***a* included in the assembled battery **301**, performs A/D conversion on the measured voltages, and supplies the converted voltages to the controller **310**. A current measurer **313** measures a current with use of the current detection resistor **307**, and supplies the measured current to the controller **310**. A switch controller **314** controls the charge control switch **302***a* and the discharge control switch **303***a* of the switcher **304**, based on the voltages inputted from the voltage detector **311** and the current inputted from the current measurer **313**.

[0088] When a voltage of any of the multiple secondary batteries **301***a* reaches the overcharge detection voltage or below, or reaches the overdischarge detection voltage or below, or when a large current flows suddenly, the switch controller **314** transmits a control signal to the switcher **304** to thereby prevent overcharging and overdischarging, and overcurrent charging and discharging. Here, for example, when the secondary battery is a lithium-ion secondary battery, the overcharge detection voltage is determined to be, for example, 4.20 V±0.05 V, and the overdischarge detection voltage is determined to be, for example, 2.4 V±0.1 V.

[0089] As the charge and discharge switches, for example, semiconductor switches such as MOSFETs are usable. In this case, parasitic diodes of the MOSFETs serve as the diodes **302***b* and **303***b*. When P-channel FETs are used as the charge and discharge switches, the switch controller **314** supplies control signals CO and DO to a gate of the charge control switch **302***a* and a gate of the discharge control switch **303***a*, respectively. When the charge control switch **302***a* and the discharge control switch **303***a* are of P-channel type, the charge control switch **302***a* and the discharge control switch **303***a* are turned on by a gate potential that is lower than a source potential by a predetermined value or more. In other words, in normal charging and discharging operations, the control signals CO and DO are set to a low level to turn on the charge control switch **302***a* and the discharge control switch **303***a*.

[0090] For example, upon overcharging or overdischarging, the control signals CO and DO are set to a high level to turn off the charge control switch **302***a* and the discharge control switch **303***a*.

[0091] A memory **317** includes a RAM and a ROM. For example, the memory **317** includes an erasable programmable read only memory (EPROM) as a nonvolatile memory. In the memory **317**, values including, without limitation, numerical values calculated by the controller **310** and a battery's internal resistance value of each of the secondary batteries **301***a* in an initial state measured in the manufacturing process stage, are stored in advance and are rewritable on an as-needed basis. Further, by storing a full charge capacity of the secondary battery **301***a*, it is possible to calculate, for example, a remaining capacity with the controller **310**.

[0092] A temperature detector **318** measures a temperature with use of the temperature detection device **308**, performs charge and discharge control upon abnormal heat generation, and performs correction in calculating the remaining capacity.

[0093] The secondary battery according to the foregoing embodiment of the present disclosure is mountable on, or usable to supply electric power to, for example, any of equipment including, without limitation, electronic equipment, an electric vehicle, an electric aircraft, and an electric power storage apparatus.

[0094] Examples of the electronic equipment include laptop personal computers, smartphones, tablet terminals, PDAs (i.e., mobile information terminals), mobile phones, wearable terminals, cordless phone handsets, hand-held video recording and playback devices, digital still cameras, electronic books, electronic dictionaries, music players, radios, headphones, game machines, navigation systems, memory cards, pacemakers, hearing aids, electric tools, electric shavers, refrigerators, air conditioners, televisions, stereos, water heaters, microwave ovens, dishwashers, washing machines, dryers, lighting equipment, toys, medical equipment, robots, road conditioners, and traffic lights.

[0095] Examples of the electric vehicle include railway vehicles, golf carts, electric carts, and electric automobiles including hybrid electric automobiles. The secondary battery is usable as a driving power source or an auxiliary power source for any of these electric vehicles. Examples of the electric power storage apparatuses include an electric power storage power source for architectural structures including residential houses, or for power generation facilities.

EXAMPLES

[0096] Examples of the present disclosure will be described according to an embodiment.

[Fabrication Method]

Example 1

[0097] As described below, a cylindrical secondary battery illustrated in FIG. **1**, etc. was fabricated. Here, a lithium-ion secondary battery was fabricated with nominal-value dimensions of 21 mm in diameter and 70 mm in length.

[0098] First, an aluminum foil having a thickness of 12 μm was prepared as the positive electrode current collector **21**A. Thereafter, a positive electrode mixture was obtained by mixing a layered lithium oxide as the positive electrode active material with a positive electrode binder and a conductive additive. The layered lithium oxide included lithium nickel cobalt aluminum oxide (NCA) having a Ni ratio of 85% or greater. The positive electrode binder included polyvinylidene difluoride. The conductive additive included a mixture of carbon black, acetylene black, and Ketjen black. A mixture ratio between the positive electrode active material, the positive electrode binder, and the conductive additive was set to 96.4:2:1.6. Thereafter, the positive electrode mixture was put into an organic solvent (N-methyl-2-pyrrolidone), following which the organic solvent was stirred to thereby prepare a positive electrode mixture slurry in paste form. Thereafter, the positive electrode mixture slurry was applied on respective predetermined regions of the two opposite surfaces of the positive electrode current collector **21**A with use of a coating apparatus, following which the applied positive electrode mixture slurry was dried to thereby form the positive electrode active material layers **21**B. Further, a coating material including PVDF copolymerized with TFE was applied on surfaces of the positive electrode current collector exposed region **212**, at respective regions adjacent to the positive electrode current collector covered region **211**. The applied coating material was dried to thereby form the insulating layer **21**C having a width of 3 mm and a thickness of 8 μm. Thereafter, the positive electrode active material layers **21**B were compression-molded with use of a roll pressing machine. The positive electrode **21** including the positive electrode current collector covered region **211** and the positive electrode current collector exposed region **212** was thus obtained. Thereafter, the positive electrode **21** was sheared to make the positive electrode current collector covered region **211** have a width of 60 mm in the W-axis direction, and to make the positive electrode current collector exposed region **212** have a width of 7 mm in the W-axis direction. Further, the positive electrode **21** was provided with a length in the L-axis direction of 1700 mm.

[0099] Further, a copper foil having a thickness of 8 μm was prepared as the negative electrode current collector **22**A. Thereafter, a negative electrode mixture was obtained by mixing the negative electrode active material with a negative electrode binder and a conductive additive. The negative electrode active material included a mixture of a carbon material and SiO. The carbon material included graphite. The negative electrode binder included polyvinylidene difluoride. The conductive additive included a mixture of carbon black, acetylene black, and Ketjen black. A mixture ratio between the negative electrode active material, the negative electrode binder, and the conductive additive was set to 96.1:2.9:1.0. Further, a mixture ratio between graphite and SiO in the negative electrode active material was set to 95:5. Thereafter, the negative electrode mixture was put into an organic solvent (N-methyl-2-pyrrolidone), following which the organic solvent was stirred to thereby prepare a negative electrode mixture slurry in paste form. Thereafter, the negative electrode mixture slurry was applied on respective predetermined regions of the two opposite surfaces of the negative electrode current collector **22**A with use of a coating apparatus, following which the applied negative electrode mixture slurry was dried to thereby form the negative electrode active material layers **22**B. Thereafter, the negative electrode active material layers **22**B were compression-molded with use of a roll pressing machine. The negative electrode **22** including the negative electrode current collector covered region **221** and the negative electrode current collector exposed region **222** was thus obtained. Thereafter, the negative electrode **22** was sheared to make the negative electrode current collector covered region **221** have a width of 62 mm in the W-axis direction, and to make the first part **222**A of the negative electrode current collector exposed region **222** have a width of 4 mm in the W-axis direction. Further, the negative electrode **22** was provided with a length in the L-axis direction of 1760 mm.

[0100] Thereafter, the positive electrode **21** and the negative electrode **22** were stacked, with the first separator member **23**A and the second separator member **23**B on the positive electrode **21** and the negative electrode **22**, respectively, to cause the positive electrode current collector exposed region **212** and the first part **222**A of the negative electrode current collector exposed region **222** to be on opposite sides to each other in the W-axis direction. The stacked body S**20** was thereby fabricated. At this time, the stacked body S**20** was fabricated not to allow the positive electrode active material layers **21**B to protrude from the negative electrode active material layers **22**B in the W-axis direction. As each of the first separator member **23**A and the second separator member **23**B, used was a polyethylene sheet having a width of 65 mm and a thickness of 14 μm. Thereafter, the stacked body S**20** was so wound in a spiral shape as to form the through hole **26**, and the fixing tape **46** was attached to the outermost wind of the stacked body S**20** thus wound.

[0101] Thereafter, the upper end face **41** and the lower end face **42** of the electrode wound body **20** were each locally bent by pressing an end of a 0.5-millimeter-thick flat plate against each of the upper end face **41** and the lower end face **42** in the Z-axis direction. The grooves **43** extending radiately in the radial directions (the R directions) from the through hole **26** were thereby formed.

[0102] Thereafter, substantially equal pressures were applied to the upper end face **41** and the lower end face **42** in substantially perpendicular directions from above and below the electrode wound body **20** at substantially the same time. The positive electrode current collector exposed region **212** and the first part **222**A of the negative electrode current collector exposed region **222** were thereby bent to make each of the upper end face **41** and the lower end face **42** into a flat surface. At this time, the positive electrode edge parts **212**E of the positive electrode current collector exposed region **212** positioned at the upper end face **41** were caused to bend toward the through hole **26** while overlapping each other, and the negative electrode edge parts **222**E of the negative electrode current collector exposed region **222** positioned at the lower end face **42** were caused to bend toward the through hole **26** while overlapping each other. Thereafter, the fan-shaped part **31** of the positive electrode current collector plate **24** was joined to the upper end face **41** by the laser welding, and the fan-shaped part **33** of the negative electrode current collector plate **25** was joined to the lower end face **42** by the laser welding.

[0103] Thereafter, the insulating tapes **53** and **54** were attached to the predetermined locations on the electrode wound body **20**, following which the band-shaped part **32** of the positive electrode current collector plate **24** was bent and inserted through the hole **12**H of the insulating plate **12**, and the band-shaped part **34** of the negative electrode current collector plate **25** was bent and inserted through the hole **13**H of the insulating plate **13**.

[0104] Thereafter, the electrode wound body **20** having been assembled in the above-described manner was placed into the outer package can **11**, following which the bottom part **11**B of the outer package can **11** and the negative electrode current collector plate **25** were welded to each other.

[0105] Thereafter, the narrow part **11**S was formed in the vicinity of the open end part **11**N of the outer package can **11**. Further, the electrolytic solution was injected into the outer package can **11**, following which the band-shaped part **32** of the positive electrode current collector plate **24** and the safety valve mechanism **30** were welded to each other.

[0106] As the electrolytic solution, used was a solution including a solvent prepared by adding fluoroethylene carbonate (FEC) and succinonitrile (SN) to a major solvent, i.e., ethylene carbonate (EC) and dimethyl carbonate (DMC), and including $LiBF_4$ and $LiPF_6$ as the electrolyte salt. In the lithium-ion secondary battery of the present Example, a content ratio (wt %) between EC, DMC, FEC, SN, $LiBF_4$, and $LiPF_6$ in the electrolytic solution was set to 12.7:56.2:12.0:1.0:1.0:17.1.

[0107] Lastly, sealing was performed with the gasket **15**, the safety valve mechanism **30**, and the battery cover **14**, through the use of the narrow part **11**S.

[0108] The secondary battery of Example 1 was thus obtained.

Example 2

[0109] In fabricating the positive electrode **21**, a coating material including PVDF copolymerized with HFP was applied, and the applied coating material was dried to thereby form the insulating layer **21**C having a width of 3 mm and a thickness of 8 μm. Except for this point, a secondary battery of Example 2 was fabricated in a manner similar to that of the secondary battery in Example 1.

Example 3

[0110] In fabricating the positive electrode **21**, a coating material including PVDF modified with acrylic acid was applied, and the applied coating material was dried to thereby form the insulating layer **21**C having a width of 3 mm and a thickness of 8 μm. Except for this point, a secondary battery of Example 3 was fabricated in a manner similar to that of the secondary battery in Example 1.

Example 4

[0111] In fabricating the positive electrode **21**, a coating material including PVDF modified with maleic acid was applied, and the applied coating material was dried to thereby form the insulating layer **21**C having a width of 3 mm and a thickness of 8 μm. Except for this point, a secondary battery of Example 4 was fabricated in a manner similar to that of the secondary battery in Example 1.

Comparative Example 1

[0112] In fabricating the positive electrode **21**, an acrylic coating material was applied, and the applied coating material was dried to thereby form the insulating layer **21**C having a width of 3 mm and a thickness of 8 μm. Except for this point, a secondary battery of Comparative example 1 was fabricated in a manner similar to that of the secondary battery in Example 1.

Comparative Examples 2 to 5

[0113] In fabricating the positive electrode **21**, a coating material including PVDF as an unmodified polymer was applied, and the applied coating material was dried to thereby form the insulating layer **21**C having a width of 3 mm and a thickness of 8 μm. Except for this point, secondary batteries of Comparative examples 2 to 5 were each fabricated in a manner similar to that of the secondary battery in Example 1.

Comparative Example 6

[0114] In fabricating the positive electrode **21**, a coating material including PVDF copolymerized with chlorotrifluoroethylene (CTFE) was applied, and the applied coating material was dried to thereby form the insulating layer **21**C having a width of 3 mm and a thickness of 8 μm. Except for this point, a secondary battery of Comparative example 6 was fabricated in a manner similar to that of the secondary battery in Example 1.

[Evaluation of Battery Characteristic]

[0115] The secondary batteries of Examples 1 to 4 and Comparative examples 1 and 6 each obtained as described above were subjected to measurements of a tensile strength and an elongation and to forced internal short circuit tests. The results are summarized in Table 1.

TABLE-US-00001 TABLE 1 Modified/ Tensile Elonga- copoly- strength tion Forced internal Material merized [N] [%] short circuit test Example 1 PVDF TFE 0.88 180.4 No short circuit Example 2 PVDF HFP 0.96 204.0 No short circuit Example 3 PVDF Acrylic 0.91 245.0 No short circuit acid Example 4 PVDF Maleic 1.30 260.0 No short circuit acid Comparative Acrylic None 1.31 1.2 Short circuit example 1 occurred Comparative PVDF None 1.40 5.6 Short circuit example 2 occurred Comparative PVDF None 0.80 14.0 Short circuit example 3 occurred Comparative PVDF None 0.71 16.4 Short circuit example 4 occurred Comparative PVDF None 0.83 40.0 Short circuit example 5 occurred Comparative PVDF CTFE 0.64 96.0 Short circuit example 6 occurred

[Tensile Strength and Elongation]

[0116] Strip-shaped samples were taken from the insulating layers **21**C used for the secondary batteries of Examples 1 to 4 and Comparative examples 1 to 6. Tensile tests were conducted on the samples in accordance with the standard of “JIS K 7161-2014 Plastics-Test method of tensile properties”, and the tensile strength [N], i.e., maximum stress observed during the test, and the elongation [%] at break of test pieces were measured. Note that a dimension of a part of each of the test pieces between grippers was 10 mm in length and 3 mm in width. The dimension of each of the test pieces including the portions gripped by the grippers was 60 mm in length. The tensile speed was 10 mm/min.

[Forced Internal Short Circuit Tests]

[0117] For the secondary batteries of Examples 1 to 4 and Comparative examples 1 to 6, tests were conducted in conformance with “Forced internal short circuit tests for battery cells” in “JIS C8714:2007, Safety tests for battery cells and assembled batteries of lithium ion storage batteries for use in portable electronic equipment”. Specifically, the electrode wound body of each of the secondary batteries was once unwound, and a nickel piece having an L-shape and a height of 0.2 mm, a width of 0.1 mm, and a side of 1 mm was sandwiched between the insulating layer **21**C covering the positive electrode current collector **21**A and the negative electrode active material layer **22**B, following which the electrode wound body was rewound. Thereafter, the electrode wound body was placed in a constant temperature bath set to a predetermined temperature, and was pressurized at a portion where the nickel piece was sandwiched. Pressurization was carried out until 800 N was reached by pressing a pressing tool at a rate of 0.1 mm/second to the portion of the electrode wound body where the nickel piece was sandwiched. Here, a case where a voltage drop of 50 m V or more was detected from an initial voltage was judged as a short circuit, and when a short circuit was observed, the test was terminated at that point in time.

[0118] As indicated in Table 1, in the forced internal short circuit tests, a short circuit occurred in all of Comparative examples 1 to 6, whereas no short circuit occurred in any of Examples 1 to 4. One reason for this may be that because the elongation was 180% or higher in Examples 1 to 4, no

crack or breakage occurred in the insulating layer **21**C even if foreign matter was present.
[0119] The results described above demonstrated that the secondary battery of the present disclosure made it possible to achieve superior safety.
[0120] Although the present disclosure has been described hereinabove with reference to some embodiments, the configuration of the present disclosure is not limited to the configurations described in relation to the embodiments, and is therefore modifiable in a variety of ways. For example, in the embodiments described above, a material including modified PVDF or copolymer PVDF is exemplified as a constituent material included in the insulating layer **21**C, but the present disclosure is not limited thereto, and other kinds of resin materials may be used.
[0121] Further, in the embodiments and Examples above, the description has been given of the secondary battery of a cylindrical type including the electrode wound body having a circular sectional shape; however, the present disclosure is not limited thereto. For example, the secondary battery may be a rectangular secondary battery including an electrode wound body having a sectional shape flattened in an elliptical shape.
[0122] Further, in the embodiments and Examples above, the description has been given of the case where the electrode reactant is lithium; however, the electrode reactant is not particularly limited. Accordingly, the electrode reactant may be another alkali metal such as sodium or potassium, or may be an alkaline earth metal such as beryllium, magnesium, or calcium, as described above. In addition, the electrode reactant may be another light metal such as aluminum.
[0123] The effects described herein are merely examples, and the effects of the present disclosure are not limited to those described herein. Accordingly, the present disclosure may achieve other effects.
[0124] The present disclosure may encompass the following embodiments.
<1>
[0125] A secondary battery including: [0126] a positive electrode current collector plate; [0127] a negative electrode current collector plate; and [0128] an electrode wound body disposed between the positive electrode current collector plate and the negative electrode current collector plate, the electrode wound body including a stacked body and having a through hole, the stacked body including a positive electrode, a negative electrode, and a separator and being wound, the through hole being provided through the electrode wound body in a height direction, in which [0129] the electrode wound body includes a first end face and a second end face, the first end face facing the positive electrode current collector plate in the height direction, the second end face facing the negative electrode current collector plate in the height direction, [0130] the positive electrode includes a positive electrode current collector, a positive electrode active material layer, and an insulating layer, the positive electrode active material layer covering a part of the positive electrode current collector, [0131] the positive electrode includes a positive electrode current collector covered region and a positive electrode current collector exposed region, the positive electrode current collector covered region being a region in which the positive electrode current collector is covered with the positive electrode active material layer, the positive electrode current collector exposed region being a region in which the positive electrode current collector is exposed without being covered with the positive electrode active material layer, [0132] all or a part of the positive electrode current collector exposed region forms the first end face and is coupled to the positive electrode current collector plate, [0133] the negative electrode includes a negative electrode current collector and a negative electrode active material layer, the negative electrode active material layer covering a part of the negative electrode current collector, [0134] the negative electrode includes a negative electrode current collector covered region and a negative electrode current collector exposed region, the negative electrode current collector covered region being a region in which the negative electrode current collector is covered with the negative electrode active material layer, the negative electrode current collector exposed region being a region in which the negative electrode current collector is exposed without being covered with the negative electrode active material layer,

[0135] all or a part of the negative electrode current collector exposed region forms the second end face and is coupled to the negative electrode current collector plate, and [0136] the insulating layer covers an opposing part, of the positive electrode current collector exposed region, that is opposed to the negative electrode active material layer with the separator interposed therebetween, and has an elongation of 180 percent or higher.
<2>
[0137] The secondary battery according to <1>, in which a length of the negative electrode active material layer in the height direction is longer than a length of the positive electrode active material layer in the height direction.
<3>
[0138] The secondary battery according to <1> or <2>, in which the insulating layer includes modified polyvinylidene difluoride (PVDF) or copolymer PVDF.
<4>
[0139] The secondary battery according to any one of <1> to <3>, in which the insulating layer includes PVDF being modified or copolymerized with tetrafluoroethylene (TFE), hexafluoropropylene (HFP), acrylic acid, or maleic acid.
<5>
[0140] The secondary battery according to any one of <1> to <4>, in which [0141] the first end face includes an edge part, of the positive electrode current collector exposed region, that is bent toward the through hole in a wound state, and [0142] the second end face includes an edge part, of the negative electrode current collector exposed region, that is bent toward the through hole in a wound state.
<6>
[0143] The secondary battery according to any one of <1> to <5>, further including: [0144] a cover part coupled to the positive electrode current collector plate; and [0145] an outer package can that contains the positive electrode current collector plate, the negative electrode current collector plate, and the electrode wound body, and is coupled to the negative electrode current collector plate, in which [0146] the outer package can includes a bottom part and a wall part, the wall part standing in the height direction along an outer edge of the bottom part and surrounding the electrode wound body, the wall part including an open end part that is open to allow the electrode wound body to be passed through the open end part, the open end part being on an opposite side to the bottom part, and [0147] the cover part covers the open end part of the outer package can.
<7>
[0148] A battery pack including: [0149] the secondary battery according to any one of <1> to <6>; [0150] a controller configured to control the secondary battery; and [0151] an outer package body that contains the secondary battery.
[0152] It should be understood that various changes and modifications to the embodiments described herein will be apparent to those skilled in the art. Such changes and modifications can be made without departing from the spirit and scope of the present subject matter and without diminishing its intended advantages. It is therefore intended that such changes and modifications be covered by the appended claims.

Claims

1. A secondary battery comprising: a positive electrode current collector plate; a negative electrode current collector plate; and an electrode wound body disposed between the positive electrode current collector plate and the negative electrode current collector plate, the electrode wound body including a stacked body and having a through hole, the stacked body including a positive electrode, a negative electrode, and a separator and being wound, the through hole being provided through the electrode wound body in a height direction, wherein the electrode wound body includes

a first end face and a second end face, the first end face facing the positive electrode current collector plate in the height direction, the second end face facing the negative electrode current collector plate in the height direction, the positive electrode includes a positive electrode current collector, a positive electrode active material layer, and an insulating layer, the positive electrode active material layer covering a part of the positive electrode current collector, the positive electrode includes a positive electrode current collector covered region and a positive electrode current collector exposed region, the positive electrode current collector covered region being a region in which the positive electrode current collector is covered with the positive electrode active material layer, the positive electrode current collector exposed region being a region in which the positive electrode current collector is exposed without being covered with the positive electrode active material layer, all or a part of the positive electrode current collector exposed region forms the first end face and is coupled to the positive electrode current collector plate, the negative electrode includes a negative electrode current collector and a negative electrode active material layer, the negative electrode active material layer covering a part of the negative electrode current collector, the negative electrode includes a negative electrode current collector covered region and a negative electrode current collector exposed region, the negative electrode current collector covered region being a region in which the negative electrode current collector is covered with the negative electrode active material layer, the negative electrode current collector exposed region being a region in which the negative electrode current collector is exposed without being covered with the negative electrode active material layer, all or a part of the negative electrode current collector exposed region forms the second end face and is coupled to the negative electrode current collector plate, and the insulating layer covers an opposing part, of the positive electrode current collector exposed region, that is opposed to the negative electrode active material layer with the separator interposed therebetween, and has an elongation of 180 percent or higher.
2. The secondary battery according to claim 1, wherein a length of the negative electrode active material layer in the height direction is longer than a length of the positive electrode active material layer in the height direction.
3. The secondary battery according to claim 1, wherein the insulating layer includes modified polyvinylidene difluoride (PVDF) or copolymer PVDF.
4. The secondary battery according to claim 3, wherein the insulating layer includes PVDF being modified or copolymerized with tetrafluoroethylene (TFE), hexafluoropropylene (HFP), acrylic acid, or maleic acid.
5. The secondary battery according to claim 1, wherein the first end face includes an edge part, of the positive electrode current collector exposed region, that is bent toward the through hole in a wound state, and the second end face includes an edge part, of the negative electrode current collector exposed region, that is bent toward the through hole in a wound state.
6. The secondary battery according to claim 1, further comprising: a cover part coupled to the positive electrode current collector plate; and an outer package can that contains the positive electrode current collector plate, the negative electrode current collector plate, and the electrode wound body, and is coupled to the negative electrode current collector plate, wherein the outer package can includes a bottom part and a wall part, the wall part standing in the height direction along an outer edge of the bottom part and surrounding the electrode wound body, the wall part including an open end part that is open to allow the electrode wound body to be passed through the open end part, the open end part being on an opposite side to the bottom part, and the cover part covers the open end part of the outer package can.
7. A battery pack comprising: the secondary battery according to claim 1; a controller configured to control the secondary battery; and an outer package body that contains the secondary battery.